# ParaBank Performance Test Analysis Using Apache JMeter

## 1. Introduction

This report evaluates the performance of the ParaBank demo banking application under loads of 1, 10, 20 and 50 virtual users. Testing was carried out with Apache JMeter 5.6.3 using a test plan built by Member 1. Each load was executed independently, and the following measures were recorded: throughput, average, minimum and maximum response time, the 95th percentile response time, and error percentage. The report compares the results across the loads and explains the changes observed.

## 2. Test Setup

The test plan contains four Thread Groups, one for each load level. Each group uses a ramp up period of 10 seconds and a duration of 20 seconds, and is configured to continue when a sampler fails. Every virtual user follows the same business scenario:

1. Home Page (preceded by a 300 ms delay)
2. Login
3. Accounts Overview
4. View Account Details
5. Transfer Funds
6. View Transactions
7. Logout

Each group includes a View Results Tree and an Aggregate Report listener, allowing both individual requests and summary statistics to be reviewed. The plan also uses HTTP Request Defaults, an HTTP Cookie Manager, and an HTTP Header Manager with the header Accept: application/json.

The original plan used a login that does not exist on ParaBank, along with outdated account numbers. As a result, Login, Accounts Overview and Transfer Funds failed on every request in the first 1 user run, giving an overall error rate of 41.67% (Figure 1).

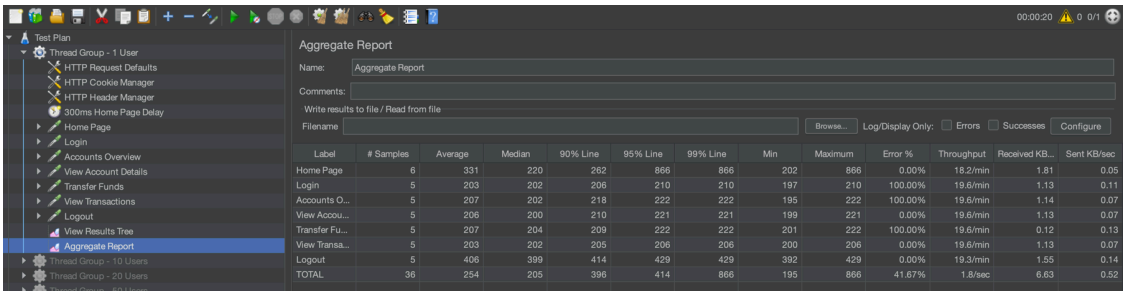



*Figure 1. First 1 user run before the login and account numbers were corrected. Login, Accounts Overview and Transfer Funds show 100% errors.*

A valid ParaBank account was registered, and the credentials and account number were updated in every request that depends on them. After this correction, the 1 user group completed with no errors. The corrected run is shown in Figure 2, with the Login and Transfer Funds request and response headers in Figures 3 and 4.

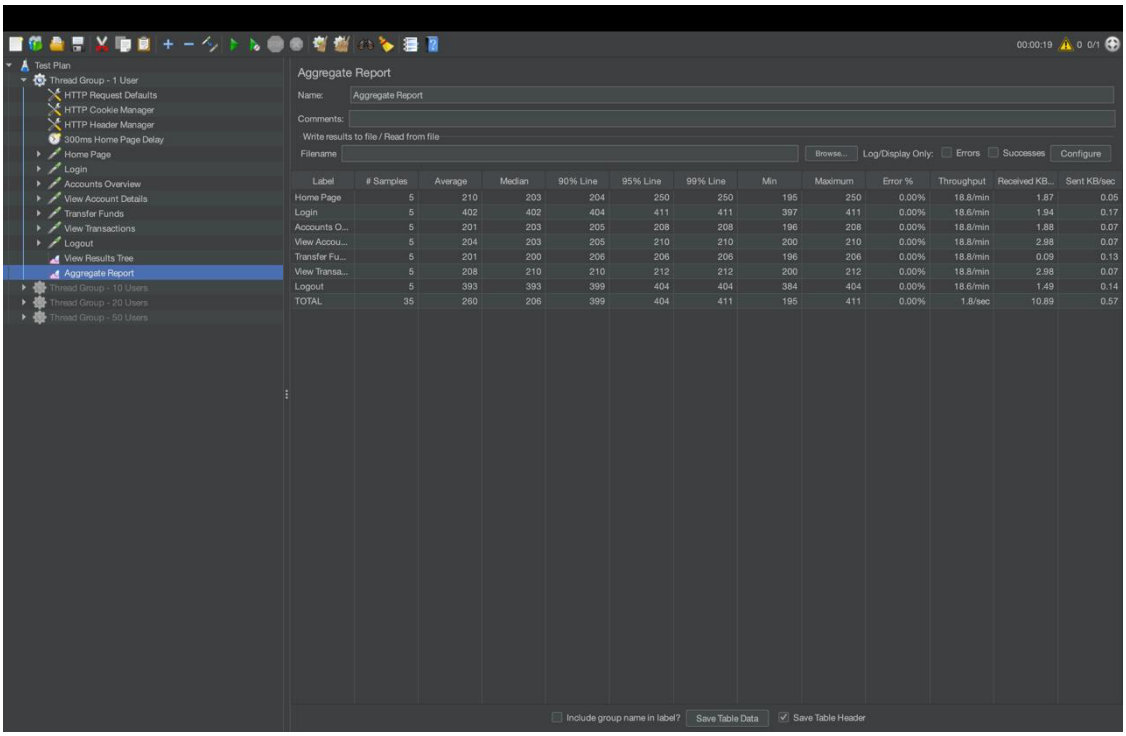
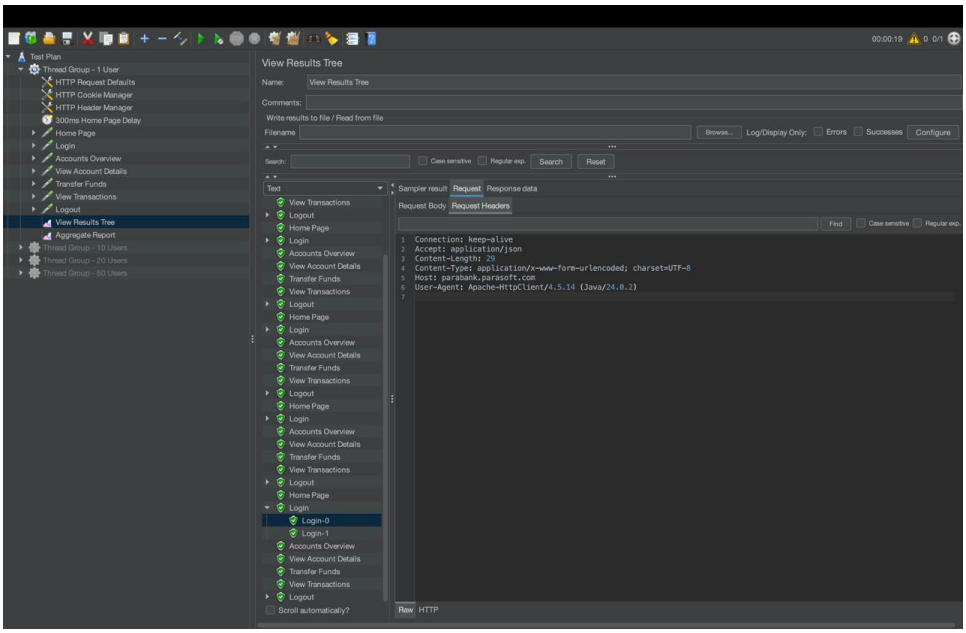



*Figure 2. 1 user, after the login and account numbers were corrected. All seven requests completed with 0.00% errors, with an average response time of 260 ms and a throughput of 1.8 requests per second.*



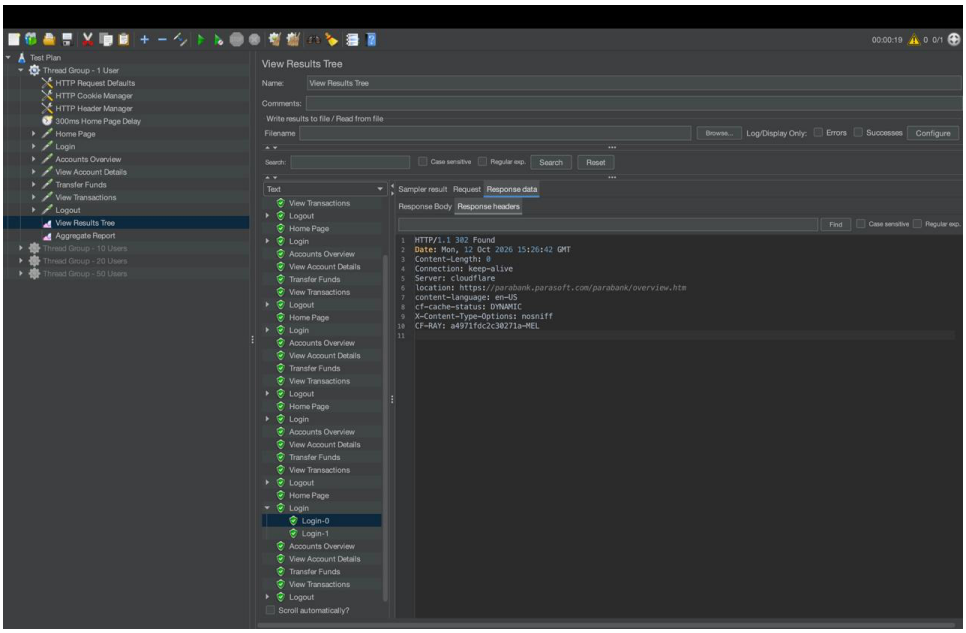




*Figure 3. Login request and response headers for the 1 user run, showing the request sent to ParaBank and the response returned by the server.*

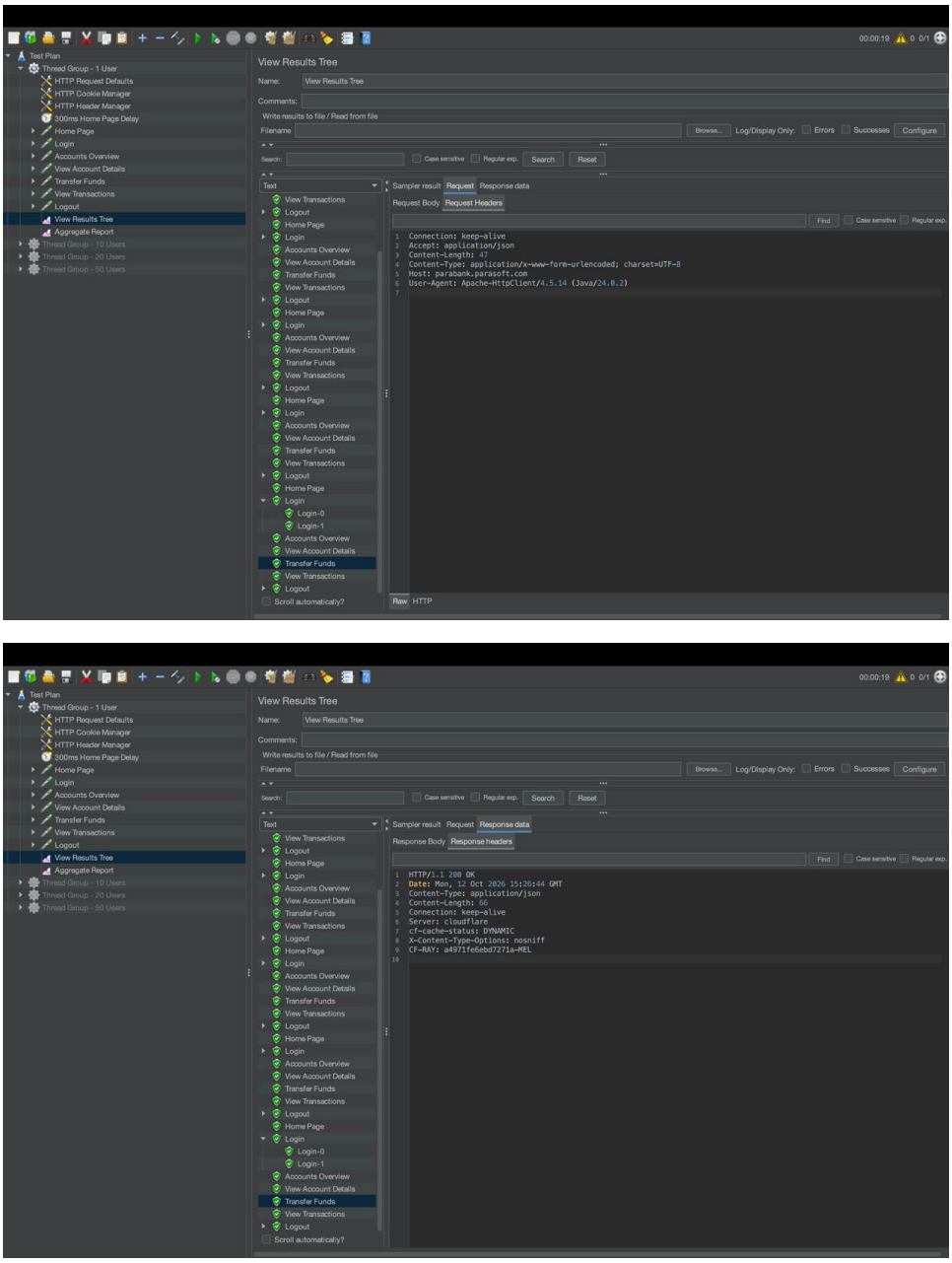



*Figure 4. Transfer Funds request and response headers for the 1 user run, showing the request parameters and the successful response from the server.*

# 3. Results

The corrected 1 user load completed without errors. The first three 10 user runs produced high error rates, which decreased as request pacing was introduced. The fourth 10 user run, with a 2000 ms Constant Timer, completed with no errors. The 20 and 50 user loads were affected by Cloudflare rate limiting, and the figures for those runs mostly reflect rejected requests. The figures below are taken from the TOTAL row of each Aggregate Report.

| Run | Samples | Req/sec | Avg | Min | Max | 95% line | Error % |
|---|---|---|---|---|---|---|---|
| 1 user | 35 | 1.8 | 260 | 195 | 411 | 404 | 0.00% |
| 10 users, run 1 | 426 | 21.6 | 55 | 5 | 766 | 370 | 83.33% |
| 10 users, run 2 | 436 | 22.1 | 47 | 5 | 594 | 227 | 85.09% |
| 10 users, run 3 (1000 ms timer) | 96 | 5.2 | 209 | 6 | 717 | 421 | 27.08% |
| 10 users, run 4 (2000 ms timer) | 56 | 3.2 | 277 | 187 | 679 | 433 | 0.00% |
| 20 users, attempt 1 (blocked from start) | 121 | 6.9 | 14 | 6 | 57 | 45 | 100.00% |
| 20 users, attempt 2 | 114 | 6.5 | 156 | 6 | 616 | 402 | 45.61% |
| 50 users | 302 | 17.1 | 14 | 6 | 111 | 40 | 100.00% |

Response times are in milliseconds. The 1 user run used only the 300 ms home page delay. Run 1 used the original plan settings. Run 2 matched the configuration of the 1 user group with no pacing. All 20 and 50 user runs used the 2000 ms Constant Timer.

In runs 1 and 2, the median response time was only 8 ms, which reflects rejected requests rather than real page responses. In run 3, the median was 204 ms, and in run 4 it was 220 ms, both close to the 1 user result.

## 3.1 10 users without pacing

With the configuration matched to the 1 user group but no pause between requests, 85.09% of requests failed (Figure 5).

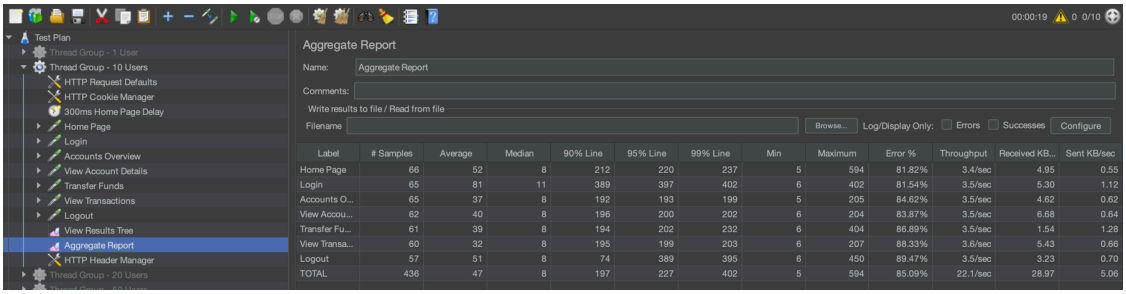



*Figure 5. 10 users, run 2, no Constant Timer. Error rate 85.09% and throughput 22.1 requests per second.*

## 3.2 10 users with a 1000 ms Constant Timer

Adding a 1000 ms Constant Timer reduced the error rate to 27.08% (Figure 6).

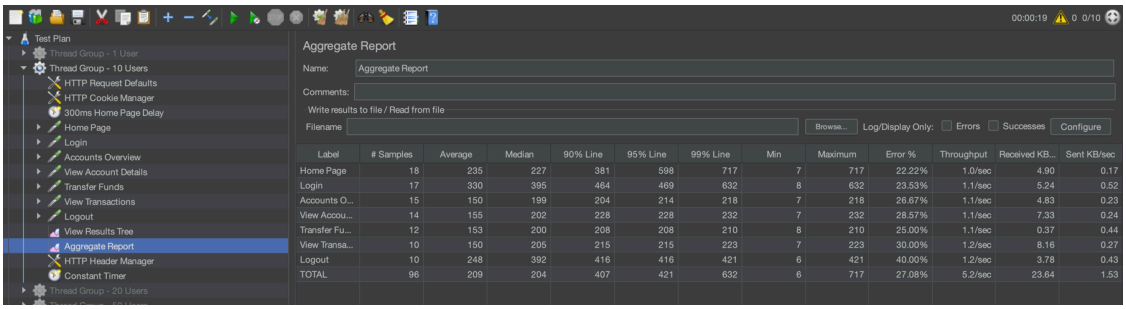



*Figure 6. 10 users, run 3, 1000 ms Constant Timer. Error rate 27.08% and throughput 5.2 requests per second.*

The View Results Tree for this run showed successful requests at the start and rejections afterwards. The response headers of a rejected request identify the source as Cloudflare (Figure 7).

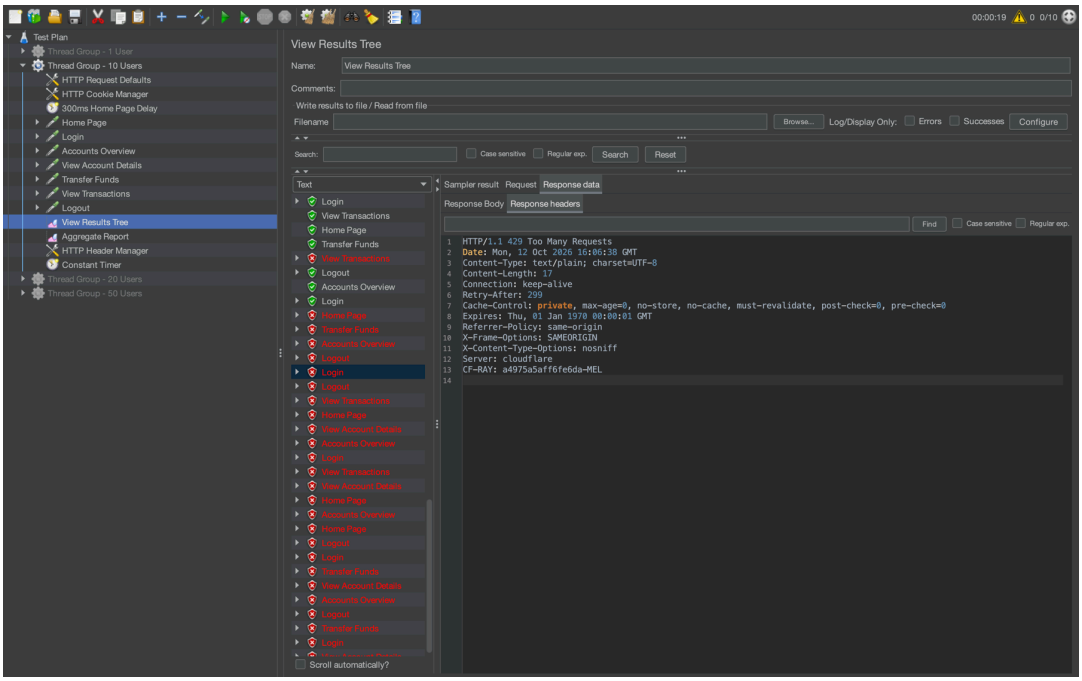



*Figure 7. Response headers of a rejected request in the 10 user run. HTTP 429 Too Many Requests, Server: cloudflare, Retry-After: 299.*

## 3.3 10 users with a 2000 ms Constant Timer

With a 2000 ms Constant Timer, all 56 requests succeeded (Figure 8).

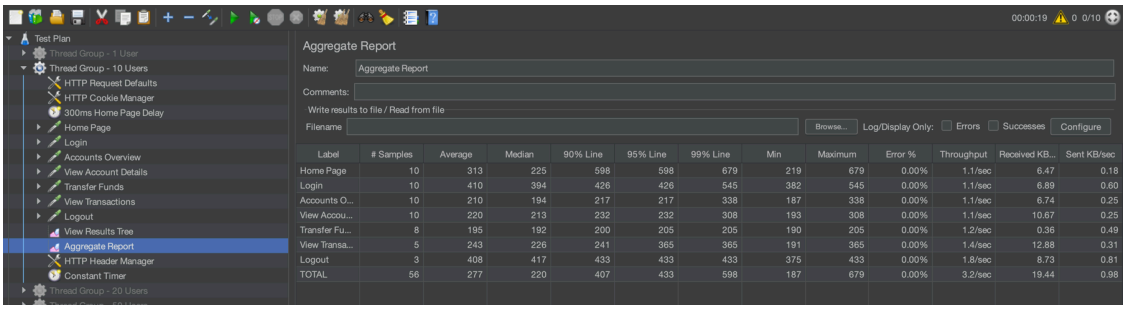



*Figure 8. 10 users, run 4, 2000 ms Constant Timer. Error rate 0.00% and throughput 3.2 requests per second.*

## 3.4 20 users

The first attempt at the 20 user load failed on every request, including the first Home Page request of each user (Figure 9).

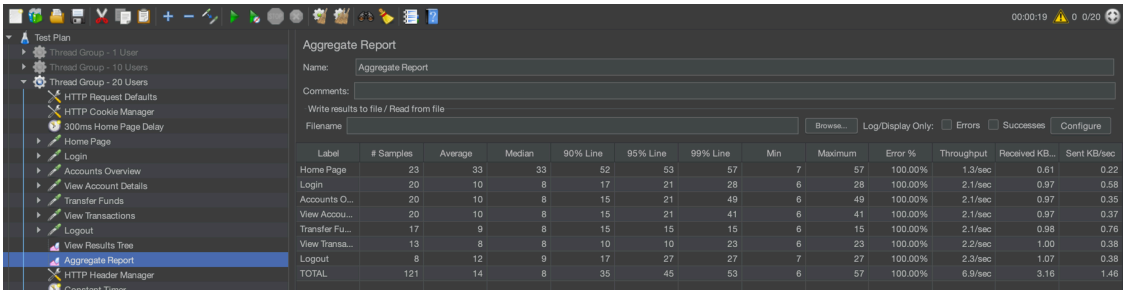



*Figure 9. 20 users, attempt 1. Error rate 100.00%, average response time 14 ms.*

The View Results Tree shows HTTP 429 responses from Cloudflare with a Retry-After value of 186 seconds (Figure 10). Because a new block starts at about 300 seconds, this suggests the block had already been active for about two minutes before the run began.

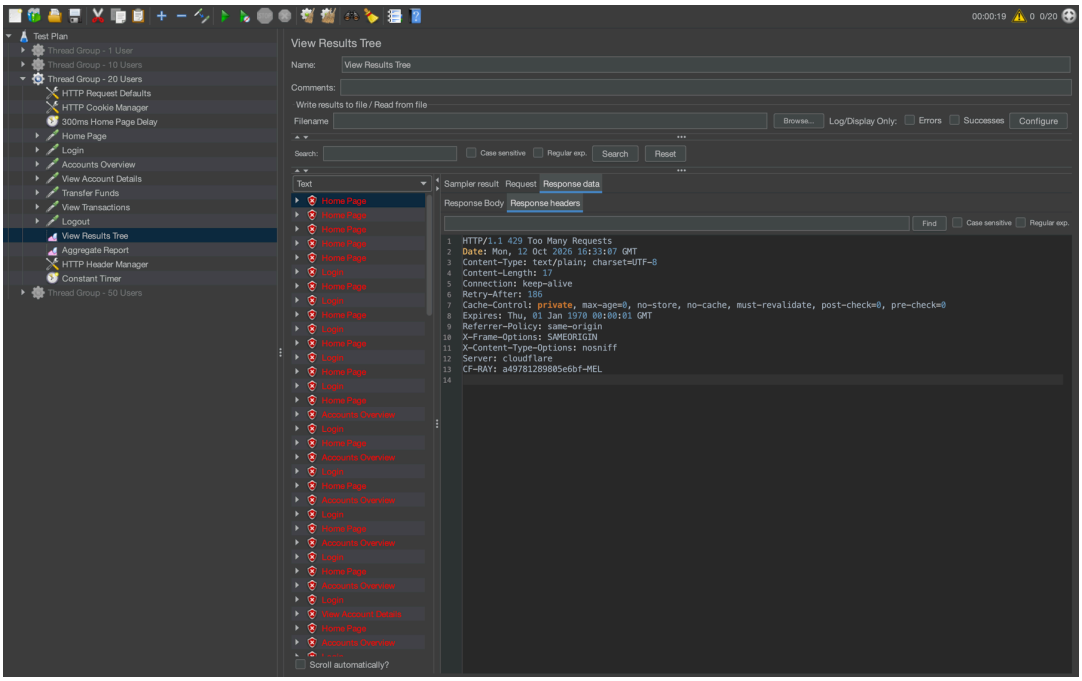



*Figure 10. 20 users, attempt 1. Response headers of a rejected request: HTTP 429, Server: cloudflare, Retry-After: 186.*

The second attempt, made after a rest period, began normally and was blocked part of the way through. The error rate was 45.61%, and it grew through the scenario: Home Page 4.76%, Login 10.00%, Accounts Overview 35.00%, View Account Details 60.00%, Transfer Funds 81.25%, and View Transactions and Logout 100.00% (Figure 11).

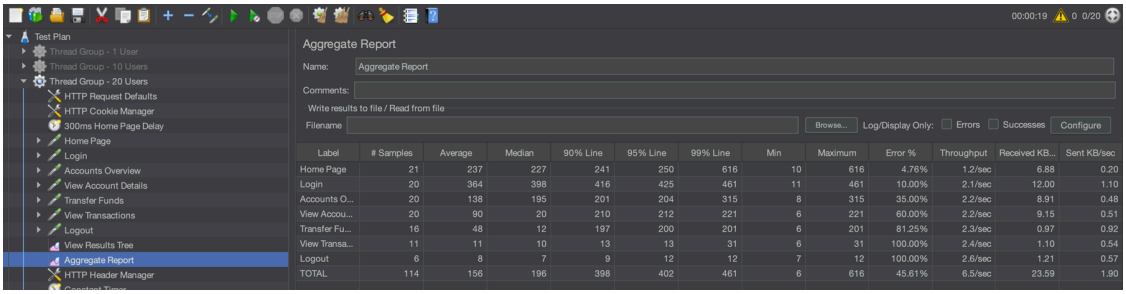



*Figure 11. 20 users, attempt 2. Error rate 45.61% and throughput 6.5 requests per second. Errors increase for the later requests in the scenario.*

In the View Results Tree, the first requests succeeded and rejections began from a Login request. That rejection carries a Retry-After value of 300, which means a new block started at that point, so this run triggered the block itself (Figure 12).

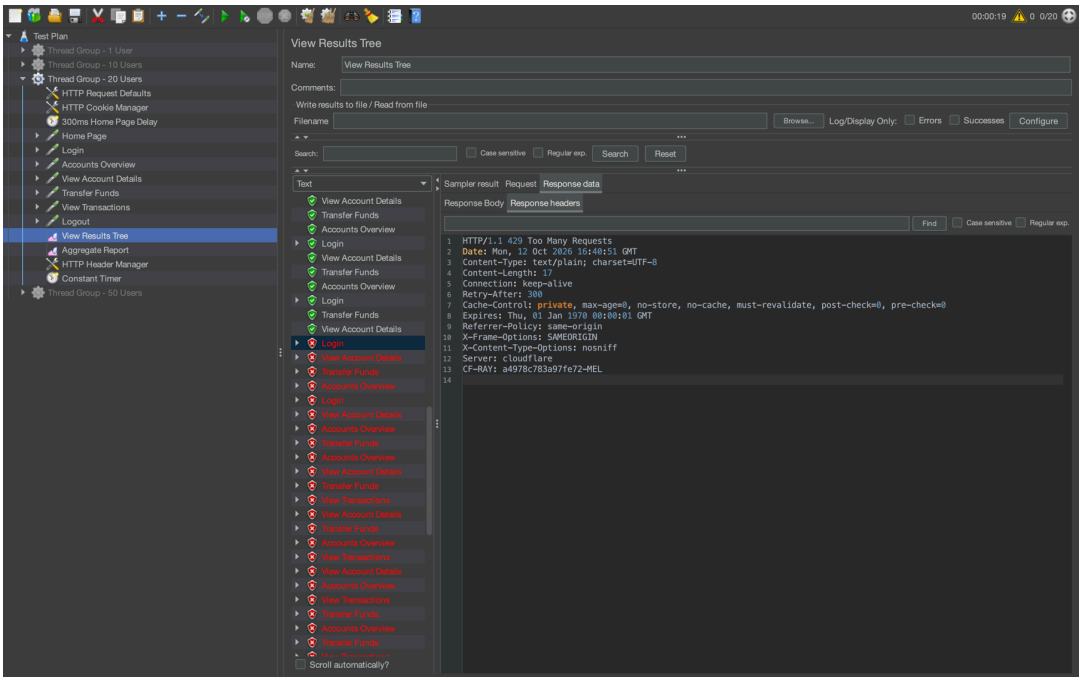



*Figure 12. 20 users, attempt 2. Successful requests followed by a rejected Login with HTTP 429, Server: cloudflare, Retry-After: 300.*

### 3.5 50 users

The 50 user run failed on every request, including Home Page, with an average response time of 14 ms and a throughput of 17.1 requests per second (Figure 13). This run began shortly after the block triggered by the second 20 user attempt, so it most likely started inside that block or at its edge. At 17 requests per second, it would also be expected to be blocked within the first seconds of a clean start.

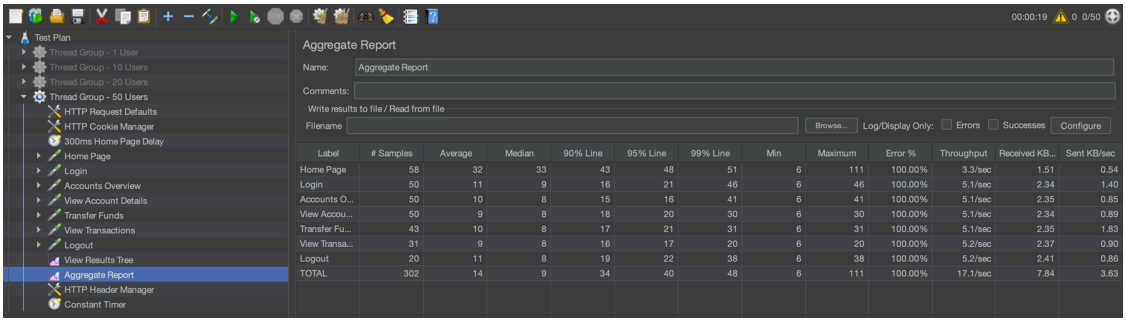



*Figure 13. 50 users. Error rate 100.00% on all requests, average response time 14 ms.*

## 4. Analysis of Errors

The errors in the 10 user, 20 user and 50 user runs did not originate from the ParaBank application. They were HTTP 429 (Too Many Requests) responses issued by Cloudflare, which sits in front of the site. The response headers showed Server: cloudflare, a CF-RAY identifier,

and a Retry-After value counting down from about 300 seconds, indicating that requests were blocked for roughly five minutes at a time.

The View Results Tree showed a consistent pattern in the 10 user runs. The first requests succeeded, after which nearly all later requests failed. This indicates that the application responded normally until the request rate exceeded the Cloudflare limit, after which further requests were rejected.

A feedback effect was also observed. A 429 response is returned in about 7 ms, so a blocked user sends its next request almost immediately. This raised the request rate to about 22 requests per second in runs 1 and 2. For this reason, the low average response time (47 to 55 ms) should not be read as good performance. It mainly reflects the speed of the rejection responses. The same applies to the first 20 user attempt and the 50 user run, where the average response time was 14 ms.

The throughput figures also help to estimate the limit. The 1 user load at 1.8 requests per second and the 10 user load at 3.2 requests per second were never blocked. The 10 user load at 5.2 requests per second and the 20 user load at 6.5 requests per second were blocked part of the way through. This suggests the limit lies somewhere between about 3 and 5 requests per second. Repeated blocked runs also appear to make later runs start inside an existing block, which is why the first 20 user attempt and the 50 user run failed from their first request.

# 5. Changes Made to the Test Plan

The following changes were made to obtain valid and comparable results:

- A valid ParaBank account was registered and used for login in every Thread Group.
- The account number was corrected in Login, View Account Details, Transfer Funds and View Transactions.
- The HTTP Cache Manager and Uniform Random Timer were removed from the 10 user group, and the same HTTP Header Manager as the 1 user group was added, so that the groups are configured consistently.
- A Constant Timer was added to the 10 user group to pace each user. At 1000 ms the error rate fell from 85.09% to 27.08%, and at 2000 ms it fell to 0.00%.
- The 20 and 50 user groups were recreated by duplicating the 10 user group, so that they have the same requests, credentials, headers and 2000 ms Constant Timer. Only the number of threads was changed.
- A rest period was left between runs, because a Cloudflare block lasts about five minutes and each run can trigger a new one.

Ideally the same pacing would be applied to all four groups so that the comparison is fair. The 1 user group was not run again with the 2000 ms timer, so this is noted as a limitation in Section 7.

## 6. Discussion

At 1 user, the application handled every request without error, with an average response time of 260 ms and a 95th percentile of 404 ms. Nearly all requests therefore completed in under about 0.4 seconds, at a throughput of 1.8 requests per second.

At 10 users with no pacing, the test degraded sharply. The error rate exceeded 83% and throughput rose to about 22 requests per second. This apparent increase in throughput is misleading, because most of those requests were 429 rejections returned almost instantly.

After pacing was added, the application responded normally. With a 2000 ms Constant Timer, the 10 user load produced no errors, an average response time of 277 ms and a 95th percentile of 433 ms. These values are only slightly higher than the 1 user result, which suggests that the application itself handles 10 concurrent users without a noticeable loss of speed. The main constraint in this test is rate limiting in front of the application, not the application itself.

At 20 users, the 2000 ms timer was no longer enough. Throughput rose to about 6.5 requests per second, the first requests succeeded, and Cloudflare began rejecting requests part of the way through, which ended with a 45.61% error rate. At 50 users, the request rate of about 17 requests per second was far above the tolerated rate, and every request was rejected. In both cases the low response times reflect rejections, not a faster application.

Taken together, the error rate rises with the number of users because the request rate rises with the number of users, and the point where errors begin matches the Cloudflare limit rather than a limit in the ParaBank application. The tests therefore show how the protection in front of ParaBank behaves under load, and they cannot show how many users the application can serve.

## 7. Limitations

- ParaBank is a public demo site protected by Cloudflare, so it cannot be tested at full load in the way an owned system can. Rate limiting stops the load before the underlying server is stressed.
- Results may vary with the time of day, network conditions, and the time elapsed since the previous run.
- The 1 user results were collected with only the 300 ms home page delay, whereas the fourth 10 user run and the 20 and 50 user runs also use a 2000 ms Constant Timer. The loads are therefore not paced identically.
- The first 20 user attempt and the 50 user run most likely began inside a block caused by an earlier run, so they show the effect of the block more than the effect of their own load.
- The test plan uses only Response Assertions, whereas the assignment brief asks for a variety of assertion types.
- The 300 ms delay is currently applied to every request in the group, whereas the brief specifies that it applies only before the home page.

# 8. Conclusion

The 1 user load and the 10 user load with a 2000 ms Constant Timer both completed with no errors, with average response times of 260 ms and 277 ms and 95th percentile times of 404 ms and 433 ms. This suggests that ParaBank performs steadily at these loads.

The 20 and 50 user loads could not be measured reliably. Cloudflare rate limiting rejected requests with HTTP 429 once the request rate went above roughly 3 to 5 requests per second, giving error rates of 45.61% at 20 users and 100.00% at 50 users. The response times recorded in these runs describe the rejection responses, not the application.

To measure the 20 and 50 user loads properly, one of two changes would be needed. The pacing between requests could be increased in proportion to the number of users, so that the total request rate stays under the limit. Alternatively, the test could be run against a local or private copy of ParaBank that has no rate limiting. In either case, the same pacing must be used for all four loads so that the results can be compared directly.